

NOTE:

1. All questions are compulsory.
2. Mark question numbers clearly on the answer sheet.

Sl. No.	Part	Question	Points
1	(a)	Starting from the time-independent Schrödinger equation, derive the stationary-state solutions for the particle-in-one dimensional box problem..	5
	(b)	Plot the wave function and the probability density for the ground state ($n = 1$) of the particle in a box.	2.5
	(c)	Plot the wave function and the probability density for the first excited state ($n = 2$) of the particle in a box.	2.5
2	(a)	An electron in a particle accelerator is moving with a speed of 3×10^7 m/s. Using the Heisenberg uncertainty principle, estimate the smallest possible uncertainty in its position. Assume the velocity is known with an accuracy of 1%.	2.5
	(b)	A car of mass 1000 kg moves at 20 m/s with an uncertainty of 0.01 m/s. Calculate its position uncertainty and compare it with the electron to comment on why quantum effects are negligible macroscopically.	2.5
	(c)	An airplane is flying at 300 m/s. How much time must elapse before a clock in the airplane and one on the ground differ by 1.00 s?	5
3	(a)	A plane diffraction grating has an overall width of 5 cm. Each transparent slit has a width of 1×10^{-4} cm and the adjacent opaque region has width 2×10^{-4} cm. Thus, the distance from the center of one slit to the center of the next slit (often called "grating element" or sometimes loosely called "distance between slits") is 3×10^{-4} cm. The wavelength of light used is $\lambda = 5.5 \times 10^{-5}$ cm. How many diffraction orders are observable?	4
	(b)	For the grating described in part (a), calculate the angular width of the central principal maximum.	3
	(c)	For the same grating, would there be any missing orders? If so, which ones?	3
4	(a)	Write down the postulates of Special Theory Relativity (STR) with its significance, and mathematically describe at least one application of STR in engineering.	4
	(b)	A particle of rest mass m is kept at rest at the origin. A constant force F starts acting on it at time $t = 0$. Show that the particle's speed always remains less than the speed of light, regardless of the magnitude of the force and/or the duration for which it is applied.	6
5	(a)	Describe how energy bands form in solids, and explain the fundamental differences between conductors, semiconductors, and insulators in terms of their band structure.	6
	(b)	What are intrinsic and extrinsic semiconductors? Explain how they differ in terms of charge carriers, band structure, and the role of impurities.	4
6	(a)	Show that in order to achieve population inversion, one needs to use a system with more than two energy levels. <i>Hint: Use the Einstein rate equations and note that for an allowed transition $B_{12} = B_{21} > 0$. Show that in steady state, even with intense pumping, the best one can reach is $N_2 = N_1$, where N_2 and N_1 are the number of atoms in energy levels 2 and 1 respectively.</i>	5

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	(b)	In a four-level laser system, the lasing transition occurs between the middle two energy levels. Explain how such a system operates.	2.5
	(c)	In the four-level laser described above, clarify why photons emitted during the rapid decay from the highest excited state to the upper lasing level (which have different frequencies) do not appear in or affect the laser output.	2.5